In [1]:
```
from pathlib import Path
import json

import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import matplotlib.image as mpimg

from IPython.display import display, Markdown

plt.rcParams["figure.dpi"] = 130
pd.set_option("display.max_colwidth", 200)
```

Mathematical framing

In this section, I study the model as a function $\hat{p}(x)$ mapping inputs to probabilities. I interpret each computation as evaluating properties of this function such as gradients, perturbations, or expectations.

1. Locate and validate the saved artefacts

In [2]:
```
# Change this if your outputs live elsewhere.
OUTDIR = Path("./analysis_outputs")

required_files = [
    "xgb_analysis_train.csv",
    "xgb_deciles.csv",
    "xgb_mean_abs_shap.csv",
    "xgb_mean_signed_shap.csv",
    "xgb_regime_signed_shap.csv",
    "xgb_case_type_signed_shap.csv",
    "xgb_interaction_candidates.csv",
    "xgb_xai_summary.json",
    "realmlp_analysis_train.csv",
    "realmlp_analysis_test.csv",
    "realmlp_fold_metrics.csv",
    "realmlp_schema_check.csv",
    "realmlp_deciles.csv",
    "realmlp_permutation_importance.csv",
    "realmlp_finite_difference_sensitivity.csv",
    "realmlp_grouped_ablation.csv",
    "realmlp_uncertainty_summary.csv",
    "realmlp_xai_summary.json",
]

missing = [f for f in required_files if not (OUTDIR / f).exists()]
if missing:
    raise FileNotFoundError(
        "The following expected artefacts were not found in "
        f"{OUTDIR.resolve()}:\n- " + "\n- ".join(missing)
    )
```

```
print("Using output directory:", OUTDIR.resolve())
print("All required exported tables and summaries were found.")
```

```
Using output directory: /Users/abhishekdey/Downloads/Predicting-Chur
n/analysis_outputs
All required exported tables and summaries were found.
```

Mathematical framing

In this section, I study the model as a function $\hat{p}(x)$ mapping inputs to probabilities. I interpret each computation as evaluating properties of this function such as gradients, perturbations, or expectations.

2. Load exported tables and summaries

In [3]:
```
xgb_train = pd.read_csv(OUTDIR / "xgb_analysis_train.csv")
xgb_dec = pd.read_csv(OUTDIR / "xgb_deciles.csv")
xgb_mean_abs_shap = pd.read_csv(OUTDIR / "xgb_mean_abs_shap.csv")
xgb_mean_signed_shap = pd.read_csv(OUTDIR / "xgb_mean_signed_shap.c
xgb_regime_signed = pd.read_csv(OUTDIR / "xgb_regime_signed_shap.cs
xgb_case_type_signed = pd.read_csv(OUTDIR / "xgb_case_type_signed_s
xgb_interactions = pd.read_csv(OUTDIR / "xgb_interaction_candidates
xgb_summary = json.loads((OUTDIR / "xgb_xai_summary.json").read_tex

real_train = pd.read_csv(OUTDIR / "realmlp_analysis_train.csv")
real_test = pd.read_csv(OUTDIR / "realmlp_analysis_test.csv")
real_fold_metrics = pd.read_csv(OUTDIR / "realmlp_fold_metrics.csv"
real_schema = pd.read_csv(OUTDIR / "realmlp_schema_check.csv")
real_dec = pd.read_csv(OUTDIR / "realmlp_deciles.csv")
real_perm = pd.read_csv(OUTDIR / "realmlp_permutation_importance.cs
real_sens = pd.read_csv(OUTDIR / "realmlp_finite_difference_sensiti
real_ablation = pd.read_csv(OUTDIR / "realmlp_grouped_ablation.csv"
real_unc = pd.read_csv(OUTDIR / "realmlp_uncertainty_summary.csv",
real_summary = json.loads((OUTDIR / "realmlp_xai_summary.json").rea

# normalise one XGBoost export depending on whether it came out as
if xgb_mean_abs_shap.shape[1] == 2 and "mean_abs_shap" in xgb_mean_
    feature_col = [c for c in xgb_mean_abs_shap.columns if c != "me
    xgb_mean_abs_shap = xgb_mean_abs_shap.rename(columns={feature_c
if xgb_mean_signed_shap.shape[1] == 2 and "mean_signed_shap" in xgb
    feature_col = [c for c in xgb_mean_signed_shap.columns if c !=
    xgb_mean_signed_shap = xgb_mean_signed_shap.rename(columns={fea

print("Loaded all exported outputs.")
print("XGB train rows:", len(xgb_train))
print("RealMLP train rows:", len(real_train))
```

```
Loaded all exported outputs.
XGB train rows: 594194
RealMLP train rows: 594194
```

Mathematical framing

In this section, I study the model as a function $\hat{p}(x)$ mapping inputs to probabilities. I interpret each computation as evaluating properties of this function such as gradients, perturbations, or expectations.

3. High-level comparison table

In [4]:
```
def _safe_float(d, key):
    val = d.get(key, np.nan)
    try:
        return float(val)
    except Exception:
        return np.nan

comparison_summary = pd.DataFrame([
    {
        "model": "XGBoost",
        "auc": _safe_float(xgb_summary, "train_auc_competition_only
        "logloss": _safe_float(xgb_summary, "train_logloss_competit
        "brier": _safe_float(xgb_summary, "train_brier_competition_
        "n_features_runtime": xgb_summary.get("n_runtime_cols", np.
        "n_feature_importance_table_rows": len(xgb_mean_abs_shap),
    },
    {
        "model": "RealMLP",
        "auc": _safe_float(real_summary, "oof_auc"),
        "logloss": _safe_float(real_summary, "oof_logloss"),
        "brier": _safe_float(real_summary, "oof_brier"),
        "n_features_runtime": real_summary.get("n_numeric_features"
        "n_feature_importance_table_rows": len(real_perm),
    },
]).sort_values("auc", ascending=False).reset_index(drop=True)

display(comparison_summary)
```

	model	auc	logloss	brier	n_features_runtime	n_feature_imp
0	XGBoost	0.933437	0.268055	0.085993	161	
1	RealMLP	0.917199	0.298696	0.095583	22	

Mathematical framing

In this section, I study the model as a function $\hat{p}(x)$ mapping inputs to probabilities. I interpret each computation as evaluating properties of this function such as gradients, perturbations, or expectations.

What to look for in the conclusion

Use this table to anchor the report quantitatively. The point is not only which model has the larger AUC, but also whether the gains are accompanied by

better log-loss and Brier score. If one model is stronger on discrimination but not on probability quality, say that explicitly.

Mathematical framing

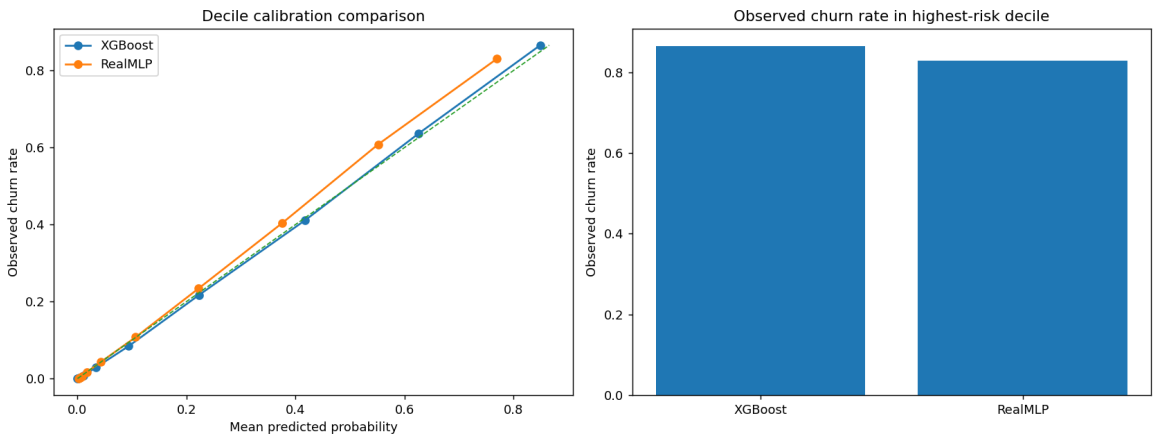
In this section, I study the model as a function $\hat{p}(x)$ mapping inputs to probabilities. I interpret each computation as evaluating properties of this function such as gradients, perturbations, or expectations.

4. Calibration and risk concentration comparison

In [5]:
```
# make the decile tables easy to compare
xgb_dec_cmp = xgb_dec.rename(columns={
    "mean_pred": "xgb_mean_pred",
    "churn_rate": "xgb_observed_rate",
    "n": "xgb_n",
    "pred_min": "xgb_pred_min",
    "pred_max": "xgb_pred_max",
})
real_dec_cmp = real_dec.rename(columns={
    "mean_pred": "real_mean_pred",
    "churn_rate": "real_observed_rate",
    "n": "real_n",
    "pred_min": "real_pred_min",
    "pred_max": "real_pred_max",
})

dec_compare = xgb_dec_cmp.merge(real_dec_cmp, on="bin", how="inner"
display(dec_compare)
```

	bin	xgb_mean_pred	xgb_observed_rate	xgb_n	xgb_pred_min	xgb_pred_
0	0	0.000052	0.000000	59420	0.000003	0.0(
1	1	0.000682	0.000236	59419	0.000110	0.00
2	2	0.003563	0.002003	59419	0.001696	0.00
3	3	0.011040	0.007927	59420	0.005920	0.01
4	4	0.033567	0.028728	59419	0.018741	0.05
5	5	0.093750	0.084165	59419	0.053925	0.14
6	6	0.223051	0.217115	59420	0.145461	0.31
7	7	0.417723	0.411215	59419	0.313006	0.52
8	8	0.625240	0.635672	59419	0.523141	0.72
9	9	0.848388	0.865012	59420	0.727878	0.99

In [6]:
```
fig, axes = plt.subplots(1, 2, figsize=(13, 5))

# calibration-like line comparison
axes[0].plot(dec_compare["xgb_mean_pred"], dec_compare["xgb_observe
axes[0].plot(dec_compare["real_mean_pred"], dec_compare["real_obser
diag_min = min(dec_compare[["xgb_mean_pred","xgb_observed_rate","re
diag_max = max(dec_compare[["xgb_mean_pred","xgb_observed_rate","re
axes[0].plot([diag_min, diag_max], [diag_min, diag_max], linestyle=
axes[0].set_title("Decile calibration comparison")
axes[0].set_xlabel("Mean predicted probability")
axes[0].set_ylabel("Observed churn rate")
axes[0].legend()

# top-decile lift / concentration comparison
lift_df = pd.DataFrame({
    "model": ["XGBoost", "RealMLP"],
    "top_decile_observed_rate": [
        dec_compare["xgb_observed_rate"].iloc[-1],
        dec_compare["real_observed_rate"].iloc[-1],
    ],
    "bottom_decile_observed_rate": [
        dec_compare["xgb_observed_rate"].iloc[0],
        dec_compare["real_observed_rate"].iloc[0],
    ],
})
axes[1].bar(lift_df["model"], lift_df["top_decile_observed_rate"])
axes[1].set_title("Observed churn rate in highest-risk decile")
axes[1].set_ylabel("Observed churn rate")

plt.tight_layout()
plt.show()
```



Mathematical framing

In this section, I study the model as a function $\hat{p}(x)$ mapping inputs to probabilities. I interpret each computation as evaluating properties of this function such as gradients, perturbations, or expectations.

What to look for in the conclusion

These are cross-model plots and should be compared directly. In the left

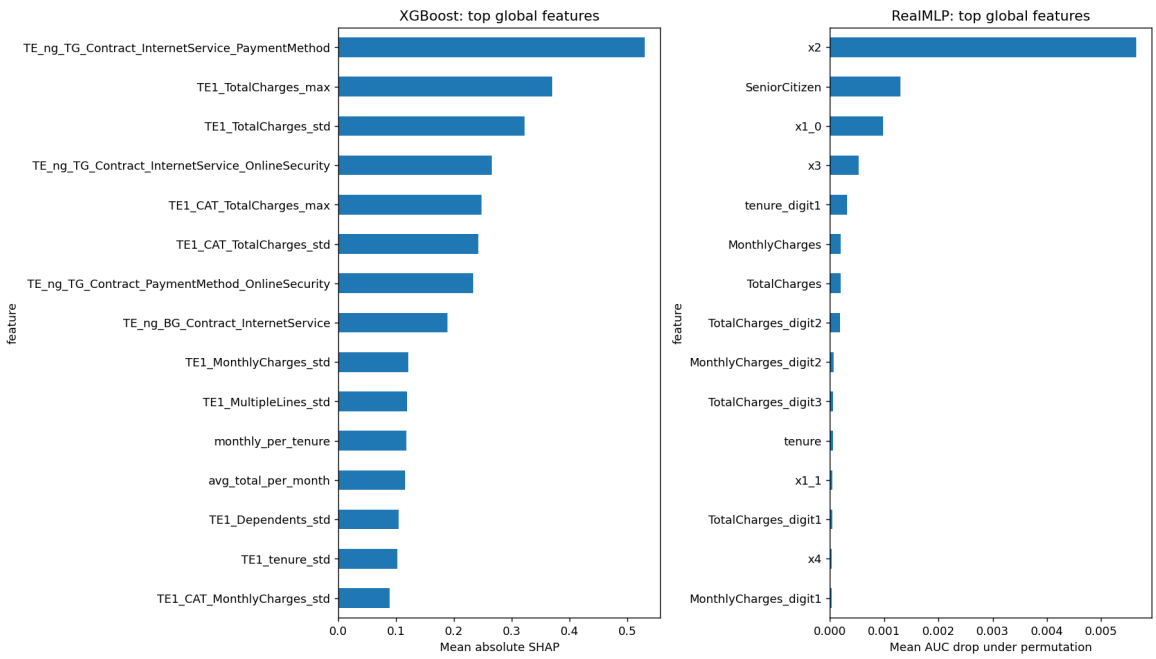
panel, the closer a model stays to the diagonal, the better its decile-level probability calibration. In the right panel, the higher the top-decile observed rate, the better the model is at concentrating true churners into the most actionable segment.

Mathematical framing

In this section, I study the model as a function $\hat{p}(x)$ mapping inputs to probabilities. I interpret each computation as evaluating properties of this function such as gradients, perturbations, or expectations.

5. Global feature reliance: ranking comparison

In [7]:
```
xgb_top = xgb_mean_abs_shap.sort_values("mean_abs_shap", ascending=
real_top = real_perm.sort_values("mean_importance", ascending=False

fig, axes = plt.subplots(1, 2, figsize=(14, 8))

xgb_top.sort_values("mean_abs_shap").plot(
    x="feature", y="mean_abs_shap", kind="barh", legend=False, ax=a
)
axes[0].set_title("XGBoost: top global features")
axes[0].set_xlabel("Mean absolute SHAP")

real_top.sort_values("mean_importance").plot(
    x="feature", y="mean_importance", kind="barh", legend=False, ax
)
axes[1].set_title("RealMLP: top global features")
axes[1].set_xlabel("Mean AUC drop under permutation")

plt.tight_layout()
plt.show()

display(pd.DataFrame({
    "xgb_top_features": xgb_top["feature"].tolist(),
    "realmlp_top_features": real_top["feature"].tolist(),
}))
```



	xgb_top_features	realmlp_top_features
0	TE_ng_TG_Contract_InternetService_PaymentMethod	x2
1	TE1_TotalCharges_max	SeniorCitizen
2	TE1_TotalCharges_std	x1_0
3	TE_ng_TG_Contract_InternetService_OnlineSecurity	x3
4	TE1_CAT_TotalCharges_max	tenure_digit1
5	TE1_CAT_TotalCharges_std	MonthlyCharges
6	TE_ng_TG_Contract_PaymentMethod_OnlineSecurity	TotalCharges
7	TE_ng_BG_Contract_InternetService	TotalCharges_digit2
8	TE1_MonthlyCharges_std	MonthlyCharges_digit2
9	TE1_MultipleLines_std	TotalCharges_digit3
10	monthly_per_tenure	tenure
11	avg_total_per_month	x1_1
12	TE1_Dependents_std	TotalCharges_digit1
13	TE1_tenure_std	x4
14	TE1_CAT_MonthlyCharges_std	MonthlyCharges_digit1

Mathematical framing

In this section, I study the model as a function $\hat{p}(x)$ mapping inputs to probabilities. I interpret each computation as evaluating properties of this function such as gradients, perturbations, or expectations.

What to look for in the conclusion

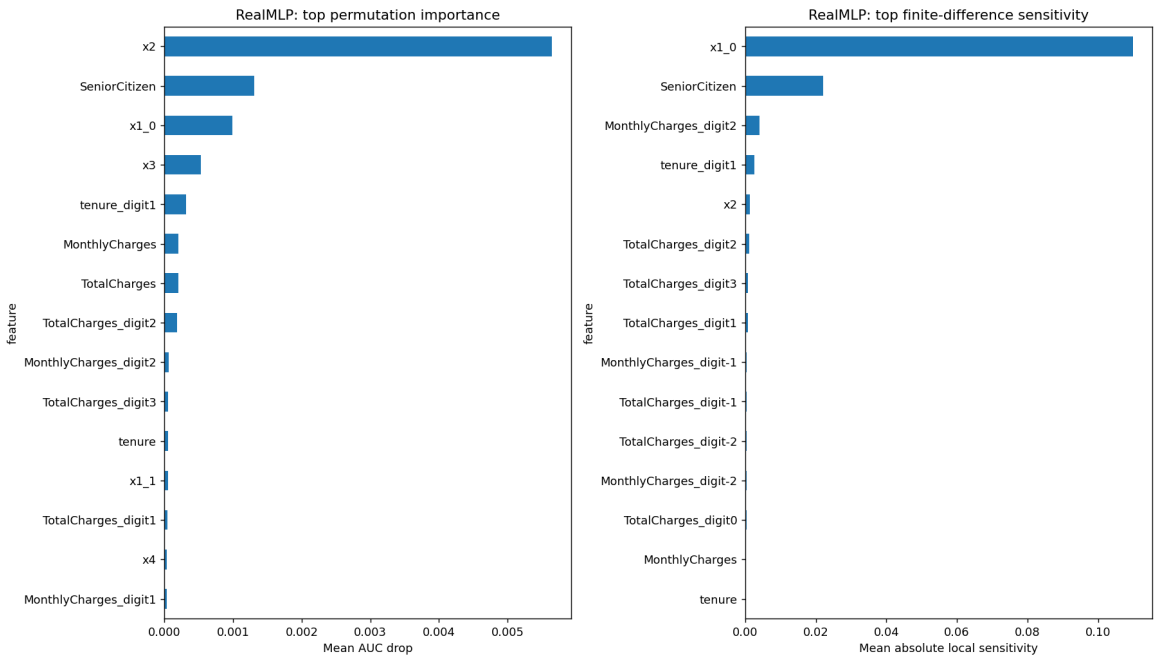
This is a cross-model comparison, but the scales are not the same, so compare **rankings and concentration**, not absolute magnitudes. If the tree model is dominated by a small number of engineered variables while the neural model spreads reliance across more inputs, that supports the claim that the two families exploit the feature space differently.

Mathematical framing

In this section, I study the model as a function $\hat{p}(x)$ mapping inputs to probabilities. I interpret each computation as evaluating properties of this function such as gradients, perturbations, or expectations.

6. RealMLP-only internal sensitivity diagnostics

In [8]:
```
fig, axes = plt.subplots(1, 2, figsize=(14, 8))

real_perm.head(15).sort_values("mean_importance").plot(
    x="feature", y="mean_importance", kind="barh", legend=False, ax
)
axes[0].set_title("RealMLP: top permutation importance")
axes[0].set_xlabel("Mean AUC drop")

real_sens.head(15).sort_values("mean_abs_sensitivity").plot(
    x="feature", y="mean_abs_sensitivity", kind="barh", legend=Fals
)
axes[1].set_title("RealMLP: top finite-difference sensitivity")
axes[1].set_xlabel("Mean absolute local sensitivity")

plt.tight_layout()
plt.show()
```



Mathematical framing

In this section, I study the model as a function $\hat{p}(x)$ mapping inputs to probabilities. I interpret each computation as evaluating properties of this function such as gradients, perturbations, or expectations.

What to look for in the conclusion

This is a **within-RealMLP** analysis and should be interpreted in isolation, not numerically compared against XGBoost. Compare the two panels to see whether the features that matter most globally are also the most locally responsive. Large permutation importance with modest sensitivity can indicate a feature that matters through broader structural effects rather than steep local gradients.

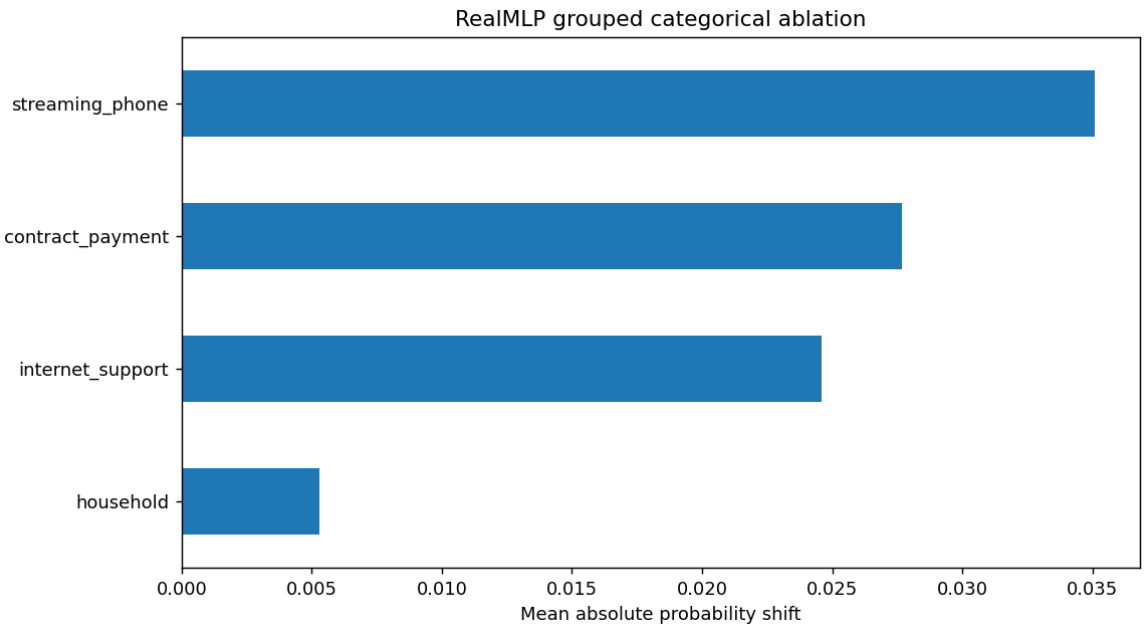
Mathematical framing

In this section, I study the model as a function $\hat{p}(x)$ mapping inputs to probabilities. I interpret each computation as evaluating properties of this function such as gradients, perturbations, or expectations.

7. RealMLP-only grouped ablation

In [9]:
```
display(real_ablation)

ax = real_ablation.sort_values("mean_abs_delta").plot(
    x="group",
    y="mean_abs_delta",
    kind="barh",
    legend=False,
    figsize=(9, 5),
)
ax.set_title("RealMLP grouped categorical ablation")
ax.set_xlabel("Mean absolute probability shift")
ax.set_ylabel("")
plt.tight_layout()
plt.show()
```

	group	mean_abs_delta	mean_signed_delta	max_abs_delta
0	streaming_phone	0.035071	0.034967	0.223998
1	contract_payment	0.027666	-0.027666	0.198302
2	internet_support	0.024606	-0.024606	0.317824
3	household	0.005301	-0.004702	0.110057



Mathematical framing

In this section, I study the model as a function $\hat{p}(x)$ mapping inputs to probabilities. I interpret each computation as evaluating properties of this function such as gradients, perturbations, or expectations.

What to look for in the conclusion

This is also a **within-RealMLP** test. It tells you which semantic subsystems the network uses most strongly. Use it to say whether the network is driven more by contractual/payment structure, service bundle structure, or demographic structure. Do not compare the raw magnitudes here against SHAP values from XGBoost.

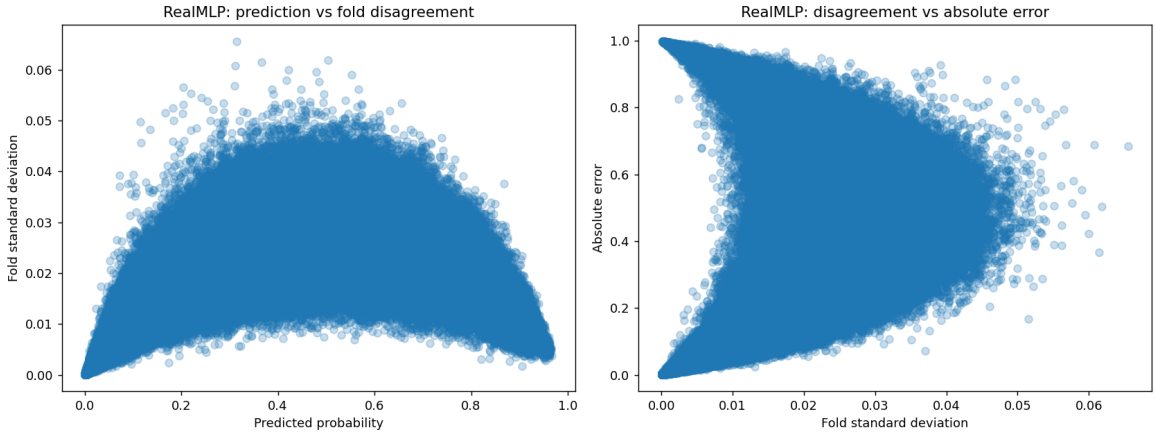
Mathematical framing

In this section, I study the model as a function $\hat{p}(x)$ mapping inputs to probabilities. I interpret each computation as evaluating properties of this function such as gradients, perturbations, or expectations.

8. RealMLP uncertainty and disagreement

In [10]:
```
fig, axes = plt.subplots(1, 2, figsize=(13, 5))

axes[0].scatter(real_train["realmlp_pred"], real_train["fold_std"],
axes[0].set_title("RealMLP: prediction vs fold disagreement")
axes[0].set_xlabel("Predicted probability")
axes[0].set_ylabel("Fold standard deviation")

axes[1].scatter(real_train["fold_std"], real_train["abs_error"], al
axes[1].set_title("RealMLP: disagreement vs absolute error")
```

```
axes[1].set_xlabel("Fold standard deviation")
axes[1].set_ylabel("Absolute error")

plt.tight_layout()
plt.show()

display(real_unc)
```



fold_std	mean_pred	observed_rate	mean_abs_error	n
(-0.0009382, 0.000666]	0.003392	0.001750	0.005128	118839
(0.000666, 0.00317]	0.012918	0.011335	0.023767	118839
(0.00317, 0.0136]	0.147528	0.151021	0.138210	118838
(0.0136, 0.0233]	0.411039	0.442700	0.354451	118839
(0.0233, 0.0656]	0.474269	0.519232	0.449057	118839

Mathematical framing

In this section, I study the model as a function $\hat{p}(x)$ mapping inputs to probabilities. I interpret each computation as evaluating properties of this function such as gradients, perturbations, or expectations.

What to look for in the conclusion

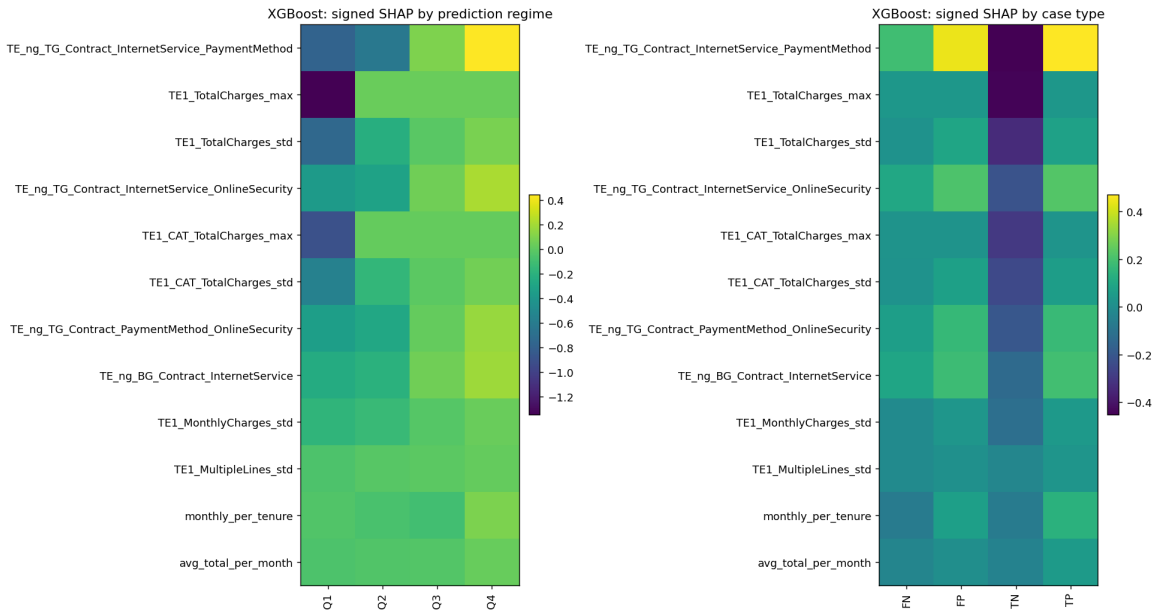
This is a **within-RealMLP** uncertainty analysis. In the left panel, look for whether disagreement is concentrated near the middle of the score range rather than the extremes. In the right panel, look for whether larger disagreement tends to coincide with larger error. If it does, then fold dispersion is acting as a meaningful uncertainty proxy rather than just random variation.

Mathematical framing

In this section, I study the model as a function $\hat{p}(x)$ mapping inputs to probabilities. I interpret each computation as evaluating properties of this function such as gradients, perturbations, or expectations.

9. XGBoost regime and case-type explanation structure

In [11]:
```
# choose a compact set of influential XGB features for visual compa
top_xgb_features = xgb_mean_abs_shap.sort_values("mean_abs_shap", a

fig, axes = plt.subplots(1, 2, figsize=(15, 8))

regime_plot = xgb_regime_signed[top_xgb_features] if set(top_xgb_fe
im0 = axes[0].imshow(regime_plot.values, aspect="auto")
axes[0].set_title("XGBoost: signed SHAP by prediction regime")
axes[0].set_yticks(range(len(regime_plot.index)))
axes[0].set_yticklabels(regime_plot.index)
axes[0].set_xticks(range(len(regime_plot.columns)))
axes[0].set_xticklabels(regime_plot.columns, rotation=90)
plt.colorbar(im0, ax=axes[0], fraction=0.046, pad=0.04)

case_plot = xgb_case_type_signed[top_xgb_features] if set(top_xgb_f
im1 = axes[1].imshow(case_plot.values, aspect="auto")
axes[1].set_title("XGBoost: signed SHAP by case type")
axes[1].set_yticks(range(len(case_plot.index)))
axes[1].set_yticklabels(case_plot.index)
axes[1].set_xticks(range(len(case_plot.columns)))
axes[1].set_xticklabels(case_plot.columns, rotation=90)
plt.colorbar(im1, ax=axes[1], fraction=0.046, pad=0.04)

plt.tight_layout()
plt.show()
```



Mathematical framing

In this section, I study the model as a function $\hat{p}(x)$ mapping inputs to probabilities. I interpret each computation as evaluating properties of this function such as gradients, perturbations, or expectations.

What to look for in the conclusion

This is an **XGBoost-only** interpretability section. It should be taken in isolation. The left panel asks how the signed contribution of important features changes as you move from low-risk to high-risk score regimes. The right panel asks whether the model is using the same features differently in true positives, false positives, true negatives, and false negatives. This is where you diagnose failure modes in the tree model rather than compare absolute attribution values against the neural model.

Mathematical framing

In this section, I study the model as a function $\hat{p}(x)$ mapping inputs to probabilities. I interpret each computation as evaluating properties of this function such as gradients, perturbations, or expectations.

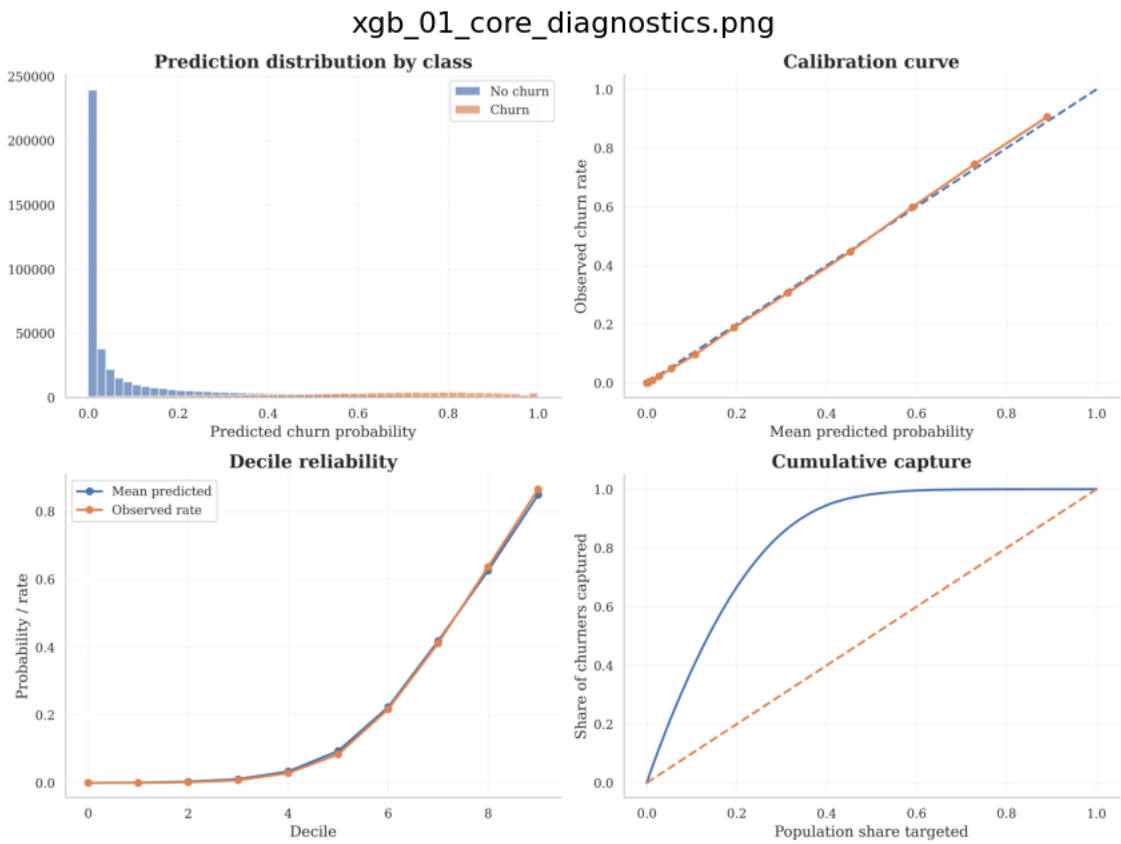
10. Previously generated model-specific figures for the appendix or report

In [12]:
```
existing_figures = [
    "xgb_01_core_diagnostics.png",
    "xgb_02_mean_abs_shap.png",
    "xgb_03_shap_beeswarm.png",
    "xgb_04_regime_signed_shap_heatmap.png",
    "xgb_05_signed_shap_by_case_type.png",
    "realmlp_01_core_diagnostics.png",
    "realmlp_02_permutation_importance.png",
    "realmlp_03_finite_difference_sensitivity.png",
    "realmlp_04_grouped_ablation.png",
    "realmlp_05_disagreement_vs_error.png",
]

available_figures = [f for f in existing_figures if (OUTDIR / f).ex
print("Available pre-rendered figures:")
for f in available_figures:
    print("-", f)
```

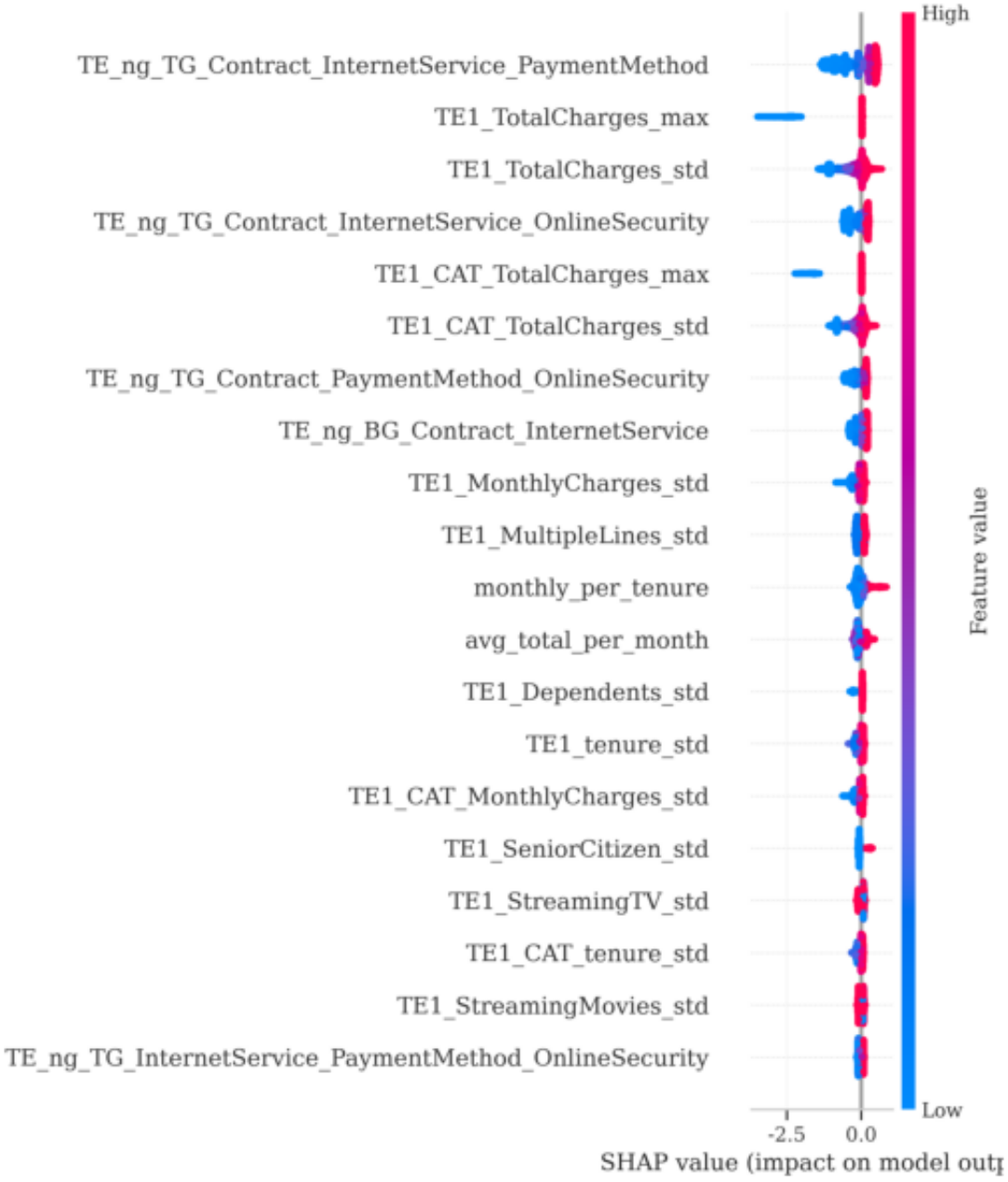
```
Available pre-rendered figures:
- xgb_01_core_diagnostics.png
- xgb_02_mean_abs_shap.png
- xgb_03_shap_beeswarm.png
- xgb_04_regime_signed_shap_heatmap.png
- xgb_05_signed_shap_by_case_type.png
- realmlp_01_core_diagnostics.png
- realmlp_02_permutation_importance.png
- realmlp_03_finite_difference_sensitivity.png
- realmlp_04_grouped_ablation.png
- realmlp_05_disagreement_vs_error.png
```

In [13]:

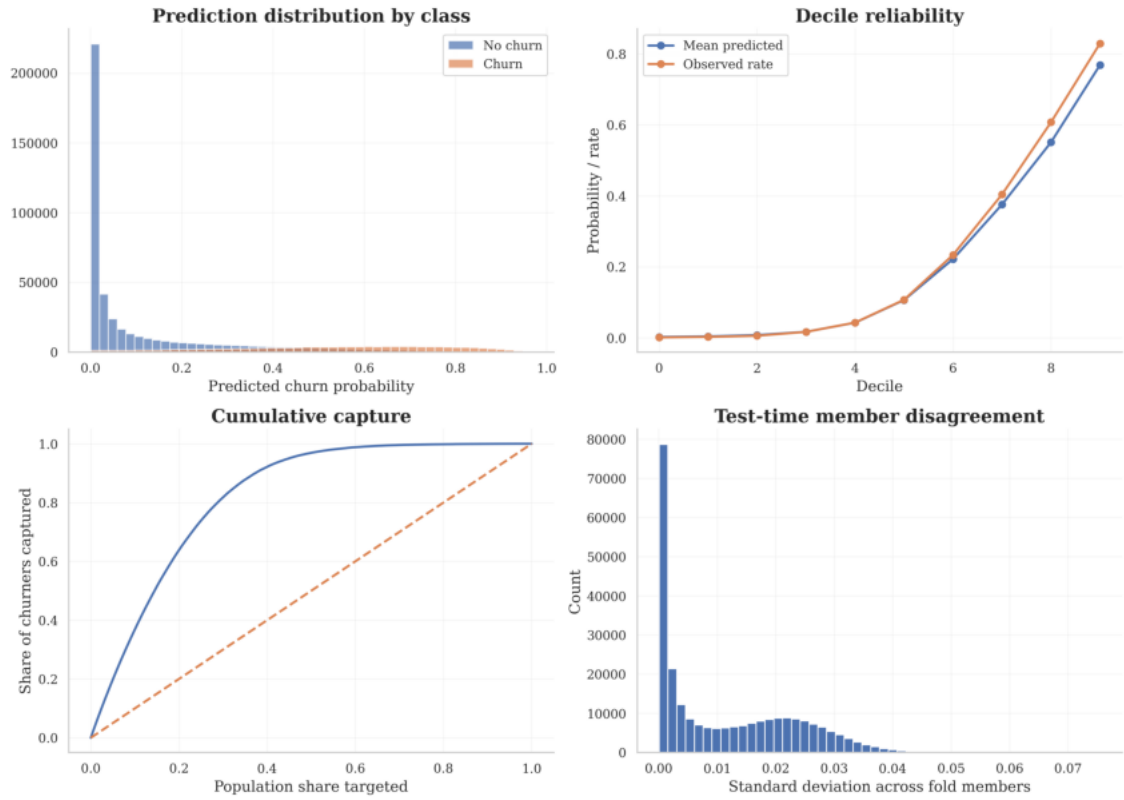
```python
# Display a small curated subset without recreating them.
to_show = [
    "xgb_01_core_diagnostics.png",
    "xgb_03_shap_beeswarm.png",
    "realmlp_01_core_diagnostics.png",
    "realmlp_05_disagreement_vs_error.png",
]

for fname in to_show:
    fpath = OUTDIR / fname
    if fpath.exists():
        img = mpimg.imread(fpath)
        plt.figure(figsize=(10, 6))
        plt.imshow(img)
        plt.axis("off")
        plt.title(fname)
        plt.show()
```



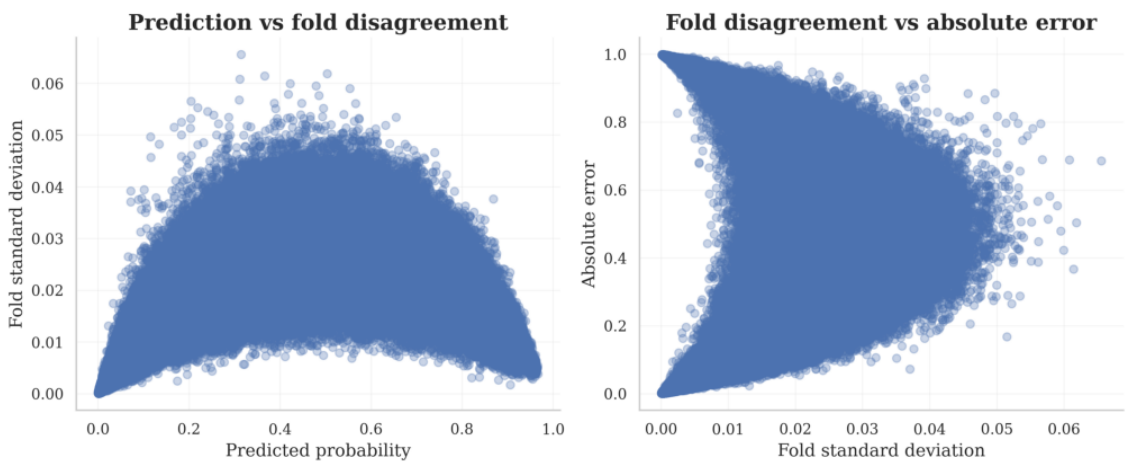
xgb_03_shap_beeswarm.png



realmlp_01_core_diagnostics.png



realmlp_05_disagreement_vs_error.png



Mathematical framing

In this section, I study the model as a function $\hat{p}(x)$ mapping inputs to probabilities. I interpret each computation as evaluating properties of this function such as gradients, perturbations, or expectations.

What to look for in the conclusion

These are the original notebook figures, kept for direct reuse in the report appendix. They are best used as supporting material after the synthesis plots above. In the report body, lead with the comparison tables and comparison plots first, then use these as model-specific evidence.

Mathematical framing

In this section, I study the model as a function $\hat{p}(x)$ mapping inputs to probabilities. I interpret each computation as evaluating properties of this function such as gradients, perturbations, or expectations.

11. Final report-ready tables

In [14]:
```
# concise report tables
report_table_metrics = comparison_summary.copy()

report_table_top_features = pd.DataFrame({
    "XGBoost top SHAP features": xgb_mean_abs_shap.sort_values("mea
    "RealMLP top permutation features": real_perm.sort_values("mean
})

# risk concentration summary
report_table_risk = pd.DataFrame([
    {
        "model": "XGBoost",
        "lowest_decile_observed_rate": dec_compare["xgb_observed_ra
        "highest_decile_observed_rate": dec_compare["xgb_observed_r
        "spread": dec_compare["xgb_observed_rate"].iloc[-1] - dec_c
    },
    {
        "model": "RealMLP",
        "lowest_decile_observed_rate": dec_compare["real_observed_r
        "highest_decile_observed_rate": dec_compare["real_observed_
        "spread": dec_compare["real_observed_rate"].iloc[-1] - dec_
    },
])

display(report_table_metrics)
display(report_table_risk)
display(report_table_top_features)
```

	model	auc	logloss	brier	n_features_runtime	n_feature_imp
0	XGBoost	0.933437	0.268055	0.085993	161	
1	RealMLP	0.917199	0.298696	0.095583	22	

	model	lowest_decile_observed_rate	highest_decile_observed_rate	spr
0	XGBoost	0.000000	0.865012	0.865
1	RealMLP	0.001094	0.829670	0.828

	XGBoost top SHAP features	RealMLP top permutation features
0	TE_ng_TG_Contract_InternetService_PaymentMethod	x2
1	TE1_TotalCharges_max	SeniorCitizen
2	TE1_TotalCharges_std	x1_0
3	TE_ng_TG_Contract_InternetService_OnlineSecurity	x3
4	TE1_CAT_TotalCharges_max	tenure_digit1
5	TE1_CAT_TotalCharges_std	MonthlyCharges
6	TE_ng_TG_Contract_PaymentMethod_OnlineSecurity	TotalCharges
7	TE_ng_BG_Contract_InternetService	TotalCharges_digit2
8	TE1_MonthlyCharges_std	MonthlyCharges_digit2
9	TE1_MultipleLines_std	TotalCharges_digit3

Mathematical framing

In this section, I study the model as a function $\hat{p}(x)$ mapping inputs to probabilities. I interpret each computation as evaluating properties of this function such as gradients, perturbations, or expectations.

What to look for in the conclusion

These are the tables to cite directly in a short report. The first gives the predictive headline numbers. The second gives the segmentation story. The third gives the interpretability headline: what each model appears to care about most.

Mathematical framing

In this section, I study the model as a function $\hat{p}(x)$ mapping inputs to probabilities. I interpret each computation as evaluating properties of this function such as gradients, perturbations, or expectations.

12. Optional export of report tables

In [15]:
```
REPORT_DIR = OUTDIR / "report_tables"
REPORT_DIR.mkdir(exist_ok=True)

report_table_metrics.to_csv(REPORT_DIR / "report_metrics.csv", inde
report_table_risk.to_csv(REPORT_DIR / "report_risk_concentration.cs
report_table_top_features.to_csv(REPORT_DIR / "report_top_features.
dec_compare.to_csv(REPORT_DIR / "report_decile_comparison.csv", ind
```

```
print("Saved report tables to:", REPORT_DIR.resolve())
```

```
Saved report tables to: /Users/abhishekdey/Downloads/Predicting-Chur
n/analysis_outputs/report_tables
```